7 Read the passage below and answer the questions that follow.

X-ray and magnetic resonance imaging (MRI) are some modern imaging techniques in medicine that uses externally placed devices to obtain diagnostic information from underneath the skin.

A modern form of X-ray tube used to obtain the internal body structure of a patient is shown in Fig. 7.1.

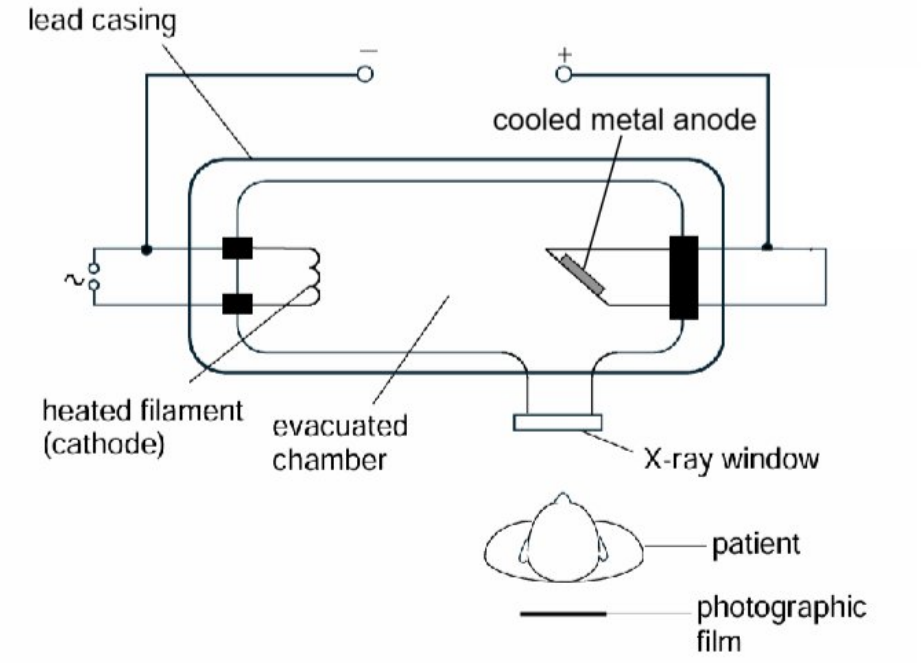


Fig. 7.1 (not to scale)

In the X-ray tube, electrons emitted from the heated filament are accelerated through a large potential difference towards the cooled metal anode, producing X-rays. The filament at the cathode is typically made of thin tungsten wire which is heated to about 2000 °C to produce electrons via thermionic emission. The X-rays produced are controlled and directed to leave the X-ray tube via a window.

As the X-ray beams pass through the patient, varying degrees of X-ray gets absorbed, depending on the composition of the body. The remaining X-rays then reach the photographic film where a contrast image of the internal body structure is obtained.

Good contrast is achieved if there is a clear difference in the blackening of the photographic film as the X-ray passes through and gets absorbed by different types of tissue in the patient. Typically, a good contrast is obtained when the ratio of the intensity of the transmitted X-rays between different body parts is at least 10.

The decrease in intensity of a beam of X-ray as it passes through matter is represented by the equation:

$$I = I_0 \, e^{(-\mu x)}$$

where

I_0 is the initial intensity,
x is the thickness of the material,
I is the transmitted intensity and μ is the absorption (attenuation) coefficient.

Table 7.1 shows the absorption (attenuation) coefficient of some matter with 30 keV X-rays.

Table 7.1

matter	μ / cm^{-1}
blood	0.41
bone	2.46
brain	0.41
muscle	0.40

(a) Suggest a reason why tungsten is used for the filament at the cathode of an X-ray tube.

.....

..... [1]

(b) Using the data in Table 7.1, determine the thickness of bone required to reduce the intensity of 30 keV X-rays to half of its initial value.

thickness = ... cm [2]

(c) A cross-section of a model arm is shown in Fig. 7.2. In an investigation into the absorption of X-ray radiation in the model arm, parallel X-ray beams of 30 keV are directed along the line MM and along the line BB.

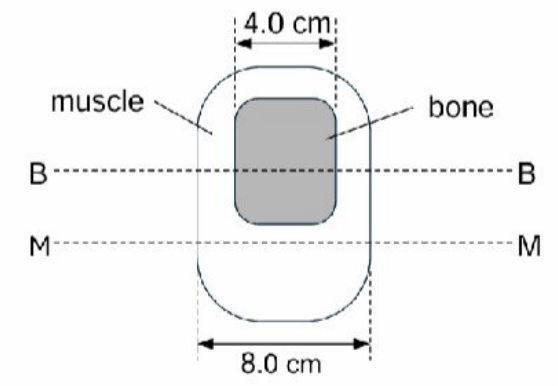


Fig.7.2

(i) Show that the ratio of

$$\frac{\text{intensity of transmitted X-ray beam from model}}{\text{intensity of incident X-ray beam on model}}$$

for a parallel X-ray beam directed along

1. the line MM is 0.041.

[1]

2. the line BB is 0.000011.

[2]

(ii) Explain whether the X-ray image obtained has good contrast.

.....

.....

..... [2]

Magnetic resonance imaging (MRI) is another imaging technique that relies on the fact that some atomic nuclei behave like tiny magnets in an external magnetic field. In MRI, it is usually the nuclei of hydrogen atoms that are studied, since hydrogen is abundant in the human body. The nucleus of a hydrogen atom consists of a single proton, which has a property called spin.

When a strong external magnetic field is applied, the spin axis of a hydrogen nucleus does not simply align directly along the external magnetic field. Instead, it rotates about the direction of the magnetic field, similar to the axis of a spinning top, as shown in Fig. 7.3. This rotational motion of the spin axis is known as precession.

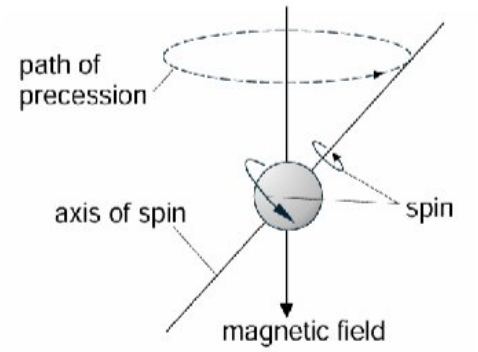


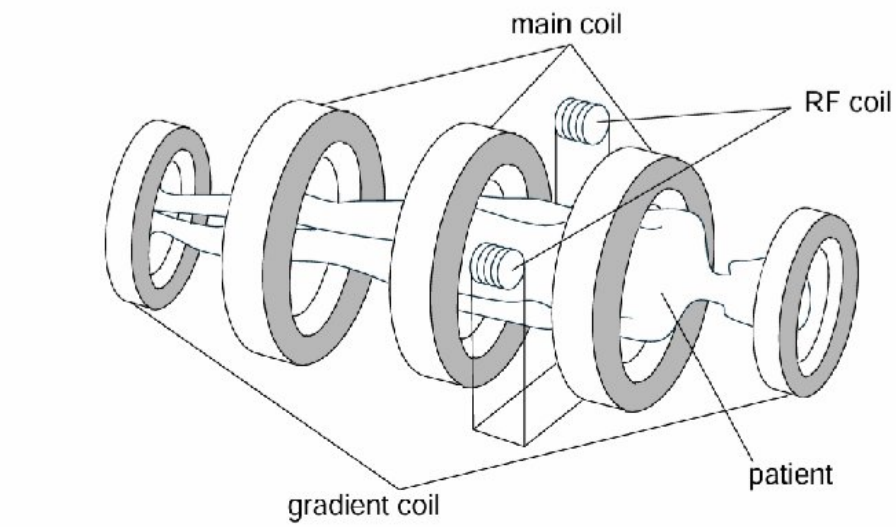
Fig. 7.3

The angular frequency of precession is called the Larmor frequency, ω_0, and depends on the individual nucleus and the magnetic flux density B_0 of the magnetic field.

$$\omega_0 = \gamma B_0$$

The quantity γ is the gyromagnetic ratio for the nucleus and is a measure of its magnetism. For hydrogen nucleus, the ratio is approximately 2.68×10^8 rad s^{-1} T^{-1}.

Fig. 7.4 shows an MRI scanner comprising three sets of coils: a main coil, gradient coils and a radio frequency (RF) coil.



The main coil is a solenoid 2.2 m long. It carries a current of 750 A to produce a strong uniform magnetic flux density of 1.5 T. To achieve superconductivity to carry the high current, the main coil is cooled using liquid helium to a temperature slightly below the boiling point of helium at 4.2 K.

The gradient coils produce additional magnetic fields that cause the magnetic flux density to vary with position across the patient. This allows signals from different regions of the body to be distinguished.

The RF coil sends RF pulses into the body, causing the hydrogen nuclei to oscillate. When the RF pulse is switched off, the nuclei relax and release energy as RF waves. The rate of relaxation depends on the type of tissue and is characterised by the spin-lattice relaxation time, *t*.

Table. 7.2 shows typical values of *t* for different tissues at a magnetic flux density of 1.5 T.

Table 7.2

matter	t / ms
blood, oxygenated	1200-1600
bone	< 100
brain, gray matter	900-1300
brain, white matter	600-800
muscle	900-1000

Different tissues can be distinguished by the different rates at which they release energy after they have been forced to oscillate.

(d) State the function of the gradient coils in the MRI scanner and suggest how this function is achieved.

.....

.....

..... [2]

(e) MRI scanning typically takes 30 minutes to more than 1 hour. Explain why MRI is particularly suitable for producing detailed images of the brain consisting of blood and brain matter compared to X-ray, despite the longer duration compared to X-ray.

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.....

..... [2]

(f) Suggest two disadvantages of using MRI.

.....

.....

.....

..... [2]

(g) Determine the number of turns in the main coil of the MRI scanner.

number of turns = ... [2]

(h) Using the Larmor frequency, determine the frequency of the pulse of the RF waves required to cause nuclear magnetic resonance of hydrogen nuclei in the MRI scanner with an external magnetic field produced by the solenoid.

frequency = ... Hz [2]

(i) During an MRI procedure, a small segment of the main coil loses its superconductivity, and the resistance of the wire suddenly increases to 0.0045 Ω. An increase in temperature of the main coil occurs, causing liquid helium to rapidly vaporise. The latent heat of vaporisation of helium is 21 kJ kg^{-1}.

Determine the initial rate of vaporisation of liquid helium.

initial rate of vaporisation = ... kg s^{-1} [2]